

INDIVIDUAL B.TECH PROJECT

IoT Accident Detection & Alert System

Real-Time Emergency Vehicle Safety using ESP32 & Telegram Messaging

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1. Executive Summary & Objectives

The Problem: Delayed medical aid post-accident drastically reduces survival chances. Unconscious or isolated drivers cannot call for emergency assistance.

Core Challenges

- Heavy dependence on eyewitness reporting in remote areas
- Inability of critical victims to initiate SOS calls manually
- Lack of precise location tracking for incoming first responders

Key Objectives

- **Automated Detection:** Continuous motion, tilt, and force monitoring via 6-axis MPU6050.
- **GPS Coordinates:** Instant location lookup via NEO-6M module upon crash recognition.
- **Instant Messaging:** Automated Telegram alert transmission to emergency contacts.
- **False Alarm Prevention:** 15-second cancellation window with local buzzer warning.

2. Key Hardware Components

ESP32

ESP32 DevKit V1

Main micro-controller unit with integrated Wi-Fi & Bluetooth, processing sensor thresholds and Wi-Fi requests.

MPU6050

MPU6050 Sensor

3-axis accelerometer and 3-axis gyroscope used to detect sudden high-g impacts and abnormal vehicle rollover angles.

NEO-6M GPS

NEO-6M GPS Module

High-precision satellite positioning unit fetching real-time latitude and longitude values for accurate geolocation.

Additional Hardware: 16x2 I2C LCD (status indication), Buzzer & Push Button (cancel switch/alarm), 5V Power Supply.

3. Accident Detection Logic & Formulas

Impact Force Calculation

Total gravitational acceleration vector is computed across 3 dimensions:

$$A_{total} = \sqrt{(A_x^2 + A_y^2 + A_z^2)}$$

Threshold: Impact triggers when $A_{total} > 2.0 g$

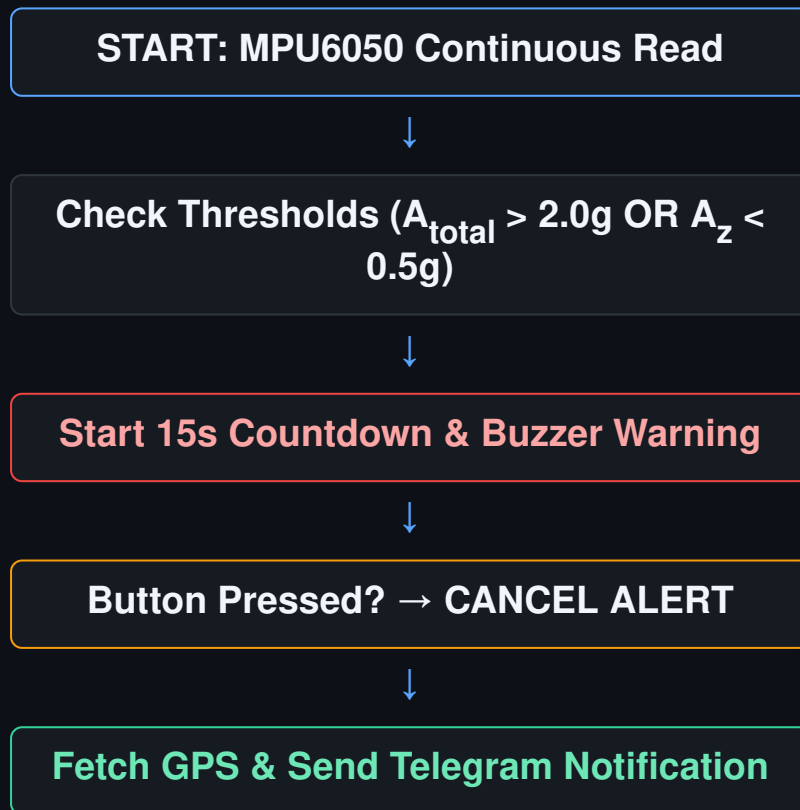
Rollover / Tilt Detection

Monitors vehicle orientation along the vertical Z-axis:

$$A_z < 0.5 g$$

Sequence Activation: If either impact force or extreme tilt is true, the 15-second cancellation countdown begins.

4. System Architecture & Workflow



Telegram SOS Alert Payload

Upon timeout expiration, ESP32 crafts and dispatches an automated HTTP payload to the configured Telegram API endpoint:

 **ACCIDENT DETECTED!** 

Vehicle ID: UP-70-EX-2026

Status: Emergency Threshold Met

Speed: 42 km/h

Google Maps Link:

<https://maps.google.com/?q=25.4358,81.8463>

5. Advantages, Future Scope & Conclusion

Key Advantages

- **Automated & Swift:** Eliminates reliance on bystander intervention.
- **Cost-Effective:** Built with readily available off-the-shelf microelectronics.
- **Precise Dispatch:** Directly transmits live GPS directions for quick medical access.

Future Enhancements

- Integration of driver vital-sign sensors (heart rate, pulse oximetry).
- On-board crash dashcam snap triggering to cloud storage.
- Fallback GSM/cellular module for regions lacking Wi-Fi coverage.

Conclusion: The developed system combines low-cost embedded hardware with high-reliability wireless protocols to solve a key vulnerability in road safety, paving the way for next-generation connected smart vehicles.